

PART 3 — INTRODUCTION

Visual-Omics Foundation Models

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● Hands-on — open the notebook

`notebooks/Part3_OmiCLIP_Introduction.ipynb`

In the next part you will:

- 1 Load the pretrained OmiCLIP model from the Loki package.
- 2 Encode ~1 000 Visium spots from a breast-cancer slide.
- 3 Inspect the cosine-similarity matrix and a UMAP of the joint space.
- 4 Run image → transcriptomics retrieval and visualize the top-k matches.

● Part 3 — Section summary

OmiCLIP = CLIP with a transcriptome on the “text” side

Vision-transformer (H&E) + expression-transformer (rank-ordered genes), contrastive InfoNCE.

Pretrained on ~2 M Visium spots, ~100 datasets

Diverse tissues, vendors, chemistries → broad transfer to downstream tasks.

The shared latent space is the primitive used by Loki & Thor in Part 4

Alignment, annotation, decomposition, retrieval, and H&E→expression all reduce to operations on this space.

From foundation model to practical analysis

Loki — A platform for visual-omics foundation model OmiCLIP

Thor — A cell-level investigation of spatial transcriptomics and histology platform

Guangyu Wang's group · Houston Methodist Research Institute

github.com/GuangyuWangLab2021/Loki · github.com/GuangyuWangLab2021/Thor

• Hands-on — Loki & Thor in practice

`notebooks/Part4_Loki_Thor_Applications.ipynb`

In the next part you will:

- 1 Align two consecutive Visium sections with Loki and inspect the alignment error.
- 2 Annotate spots with a TCGA bulk reference and with a custom marker panel.
- 3 Decompose spots into cell-type proportions using a single-cell breast atlas.
- 4 Retrieve transcriptomes from an H&E region of interest.
- 5 Run Thor on a TCGA H&E slide; correlate predicted ERBB2 with clinical HER2 status.

● Loki at a glance

Open-source Python package – github.com/GuangyuWangLab2021/Loki

Five core modules. Each takes scanpy/AnnData in and returns AnnData out — minimal friction with existing pipelines.

loki.align

Visium ↔ Visium tissue alignment via OmiCLIP image features.

loki.annotate

Bulk-RNA-seq and marker-gene annotation in the shared latent space.

loki.decompose

Cell-type decomposition by NNLS on reference centroids.

loki.retrieve

Image ↔ transcriptomics retrieval (per-spot and slide-scale).

loki.models

Convenience loaders for OmiCLIP checkpoints.

Loki #1 — Tissue alignment

- Problem — consecutive Visium sections don't share spot coordinates: rotation, translation, deformation, missing tissue.
- Use OmiCLIP image embeddings as a modality-invariant feature space; match spots by cosine similarity.
- Fit affine / thin-plate-spline warp. Optional MNN refinement.
- Three feature modes — image (robust across patients), expression (within-patient), joint.

Mouse-brain alignment benchmark

Method	Mean err. (µm)
PASTE	79
STalign	51
Loki (image)	38

Use 'tps' for tumours with significant deformation.

```
from loki.align import align_sections
result = align_sections(secA, secB, encoder=model, feature='image', transform='affine', refine_with_mnn=True)
```

● Loki #2 — Annotation by bulk RNA-seq

- Input: any bulk RNA-seq reference matrix (TCGA subtypes, GTEx tissues, in-house).
- Push every reference column through the expression encoder once — get a 512-d reference per label.
- Assign each Visium spot to the closest reference by cosine similarity.
- Gene overlap doesn't need to be 1-to-1 — only the gene tokens matter.
- No retraining, no per-dataset hyper-parameter tuning.

Bulk reference

TCGA-BRCA Luminal A / B / HER2+ / Basal



OmiCLIP expression encoder

→ 512-d embedding per subtype



Per-spot label

argmax cosine similarity
+ confidence score

● Loki #3 — Annotation by marker-gene panel

- When no bulk reference is available, supply a dictionary { cell-type → [marker genes] }.
- Each list is converted to a 'pseudo-expression' token sequence (uniform expression on the markers).
- Encode → reference embedding per panel → nearest match per spot.
- Works with as few as 3–5 markers — great for exploratory analysis on new tissues.

```
marker_panels = {  
    'Tumor epithelium' : ['ERBB2', 'EPCAM', 'KRT8', 'KRT18', 'CDH1'],  
    'T cells'          : ['CD3D', 'CD3E', 'CD8A', 'CD4', 'GZMB'],  
    'Stromal/CAF'      : ['COL1A1', 'COL3A1', 'FAP', 'ACTA2', 'PDGFRB'],  
    'Endothelium'      : ['PECAM1', 'VWF', 'CDH5', 'CLDN5'],  
    'Macrophages'      : ['CD68', 'CD163', 'C1QA', 'C1QB', 'AIF1'],  
}  
annot = marker_annotate(adata, marker_panels, encoder=model, tokenize=tokenize_expression)
```

● Loki #4 — Cell-type decomposition

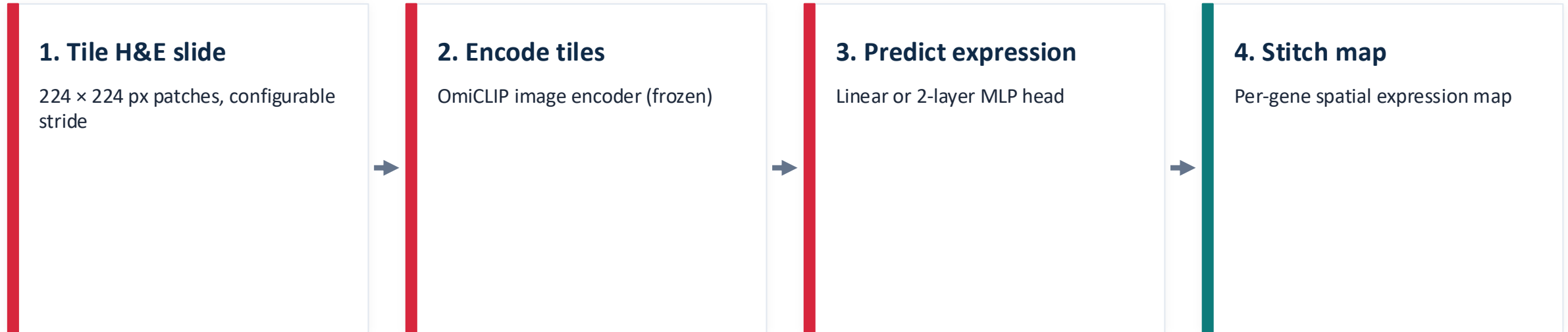
- A Visium spot $\approx$ a mixture of 1–20 cells.
- Reference embeddings come from any annotated single-cell atlas.
- Non-negative least squares in 512-d (closed-form, < 1 ms / spot).
- Returns per-spot cell-type proportions $\rightarrow$ standard `sc.pl.spatial`.
- Faster and more robust than gene-space methods because we work in a denoised low-dimensional space.

● Loki #5 — Cross-modal retrieval

- Build an index of Visium spots → expression embeddings.
- Query with any H&E patch — Visium or not (any pathology slide).
- Return nearest spots and their gene-level expression profiles.
- Use case: pathologist circles a region of interest on a slide; system retrieves matching transcriptomic profiles from a curated atlas, with top upregulated genes.

Turns any tissue atlas into a queryable resource for everyday pathology work.

• Thor — predicting gene expression from H&E alone



The regression head is trained once on ~2 M Visium spots and then applied to any H&E slide.

⇒ Bring molecular maps to TCGA, biobank cohorts, and routine pathology archives — slides where Visium was never run.

● Thor — what works, what doesn't

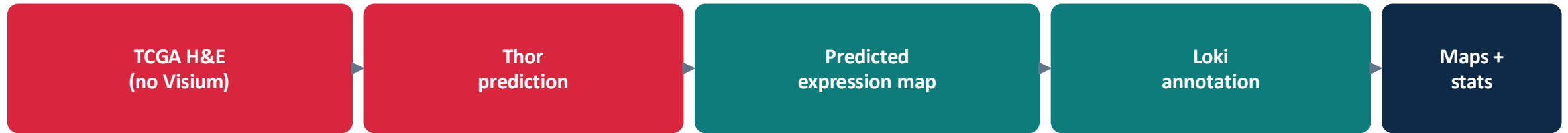
✓ Works well

- Highly expressed lineage / compartment genes (KRT8, CD3D, COL1A1, ERBB2, ...).
- Hallmark / pathway-level signals.
- Spatial domains with clear morphological correlates.
- Per-gene Pearson $r \approx 0.4$ – 0.7 for the top decile of HVGs.

✗ Be careful

- Lowly expressed TFs, ligands, cytokines — high noise.
- Rare programs without obvious morphological correlate.
- Extreme stain variability (use Macenko-style normalization first).
- Per-gene r near zero in the lowest decile — always inspect the gene list.

● End-to-end case — TCGA-BRCA H&E slide



Outputs we get for free

- Per-gene spatial expression maps (ERBB2, ESR1, MKI67, CD3D, COL1A1, ...).
- Inferred tissue compartments (tumour / stroma / immune / vasculature) overlaid on the slide.
- Surrogate HER2 / ER / PR scoring correlated with the clinical metadata table.
- Slide-level summary embedding for cohort-scale comparisons.

● Practical advice

Always inspect a UMAP first

Colour by modality. If image and expression embeddings are separated, something is off upstream.

For decomposition, invest in the reference

The reference single-cell atlas matters more than the decomposition method.

For Thor, validate on held-out Visium

Compute per-gene Pearson r ; trust the top decile, treat the bottom decile as noise.

Start small

The base OmiCLIP checkpoint is enough for most tasks. Embeddings are cheap to compute.

GET STARTED

● Resources

Loki

github.com/GuangyuWangLab2021/Loki

Thor

github.com/GuangyuWangLab2021/Thor

Q & A

Thank you!

github.com/GuangyuWangLab2021